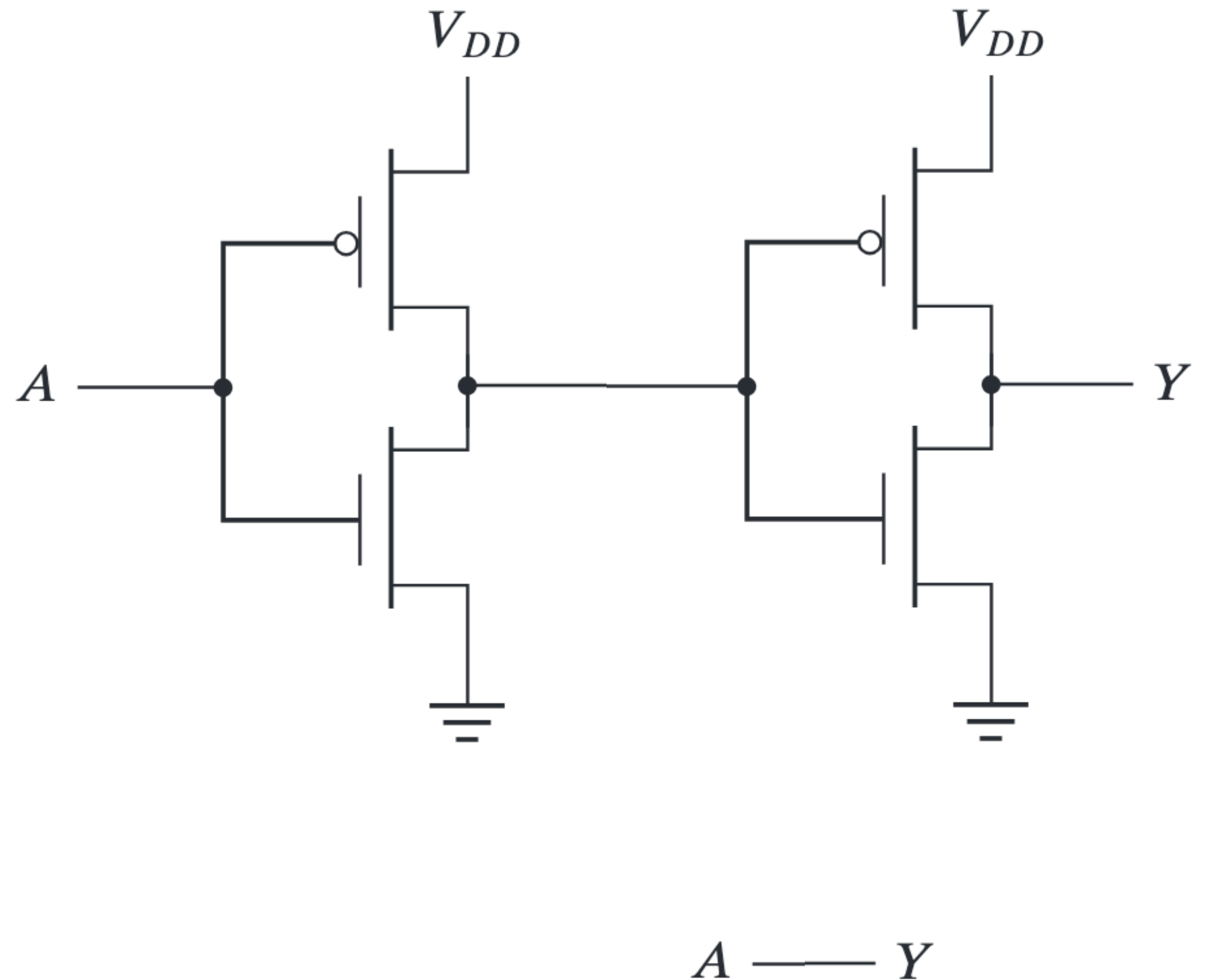


Boolean Algebra and Logic Gates

Abhilash Jindal

Introduction

- Finding simpler and cheaper, but equivalent, realisations of a circuit can reduce overall cost
- Boolean algebra to optimize simple circuits
- $Y = (A')' = A$



Binary Operators

Closure. $\forall a, b \in S : \exists c \in S : a \square b = c$

- Commutative: $\forall x, y \in S : x \square y = y \square x$
- Associative: $\forall x, y \in S : (x \square y) \square z = x \square (y \square z)$
- Identity element: $\exists e \in S : \forall x \in S : e \square x = x \square e = x$
- Inverses: $\forall x \in S : \exists y \in S : x \square y = e$
- Distributive: $\forall x, y, z \in S : x \square (y \diamond z) = (x \square y) \diamond (x \square z)$

$(B, +, *)$ is a Boolean Algebra

Huntington Postulates

- Set of elements B and two binary operators $+$ and $*$ (or $\cdot$)
 - B is closed on $+$ and $*$
 - 0 is identity with respect to $+$ and 1 is identity with respect to $*$
 $0 + x = x + 0 = x$ and $x * 1 = 1 * x = x$
 - Both $+$ and $*$ are commutative
 - $*$ distributes over $+$ and $+$ distributes over $*$
 $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and $x + (y \cdot z) = (x + y) \cdot (x + z)$
 - $\forall x \in B$, there exists a complement $x' \in B$ such that $x + x' = 1$ and $x * x' = 0$

Is $(\mathbb{R}, +, *)$ a Boolean Algebra?

Huntington Postulates

- Set of elements B and two binary operators $+$ and $*$ (or $\cdot$)
 - B is closed on $+$ and $*$
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 - $\forall x \in B$, there exists a complement $x' \in B$ such that $x + x' = 1$ and $x * x' = 0$
- + does not distributes over * !
- + and * do not have inverses.
- and / are not defined!
- Complements do not exist

Is $(\mathcal{P}(\{a, b\}), \cup, \cap)$ a Boolean Algebra?

Huntington Postulates

0 is $\{\}$. 1 is $\{a, b\}$

- B is closed on $+$ and $*$
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x' is $\{a, b\} \setminus x$

Theorems and properties

- $x + x = x$

- $x + x =$

- $(x + x) * 1$ $y * 1 = y$

Identity

- $(x + x) * (x + x')$ $(x + x') = 1$

Complement

- $x + x * x'$ $(x+y)*(x+z) = x + (y * z)$

+ Distributes over *

- $x + 0$ $x * x' = 0$

Complement

- x $y + 0 = y$

Identity

Theorems and properties

- $x + 1 = 1$

- $x + 1 =$

- $1 * (x + 1)$

$$1 * y = y$$

Identity

- $(x + x') * (x + 1)$

$$(x + x') = 1$$

Complement

- $x + x'^*1$

$$(x+y)*(x+z) = x + (y * z)$$

+ Distributes over *

- $x + x'$

$$x'^*1 = x'$$

Identity

- 1

$$x + x' = 1$$

Complement

Duality of Boolean Algebra

Huntington Postulates

- Set of elements B and two binary operators $+$ and $*$ (or $\cdot$)
 - B is closed on $+$ and $*$
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Swap $+$ and $*$. Swap 0 and 1

Is $(\mathcal{P}(\{a, b\}), \cap, \cup)$ a Boolean Algebra?

Huntington Postulates

0 is $\{a, b\}$. 1 is $\{\}$

- B is closed on $+$ and $*$
- 0 is identity with respect to $+$ and 1 is identity with respect to $*$
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x' is $\{a, b\} \setminus x$

Dual Theorems

Swap + and *. Swap 0 and 1

(a) $x + 0 = x$

(b) $x \cdot 1 = x$

(a) $x + x' = 1$

(b) $x \cdot x' = 0$

(a) $x + x = x$

(b) $x \cdot x = x$

(a) $x + 1 = 1$

(b) $x \cdot 0 = 0$

$(x')' = x$

(a) $x + y = y + x$

(b) $xy = yx$

(a) $x + (y + z) = (x + y) + z$

(b) $x(yz) = (xy)z$

(a) $x(y + z) = xy + xz$

(b) $x + yz = (x + y)(x + z)$

(a) $(x + y)' = x'y'$

(b) $(xy)' = x' + y'$

(a) $x + xy = x$

(b) $x(x + y) = x$

Dual theorem proofs

Swap + and *. Swap 0 and 1

- $x * x = x$

- $x * x =$

- $(x * x) + 0$ $y + 0 = y$

- $(x * x) + (x * x')$ $(x * x') = 0$

- $x * (x + x')$ $(x * y) + (x * z) = x * (y + z)$

- $x * 1$ $x + x' = 1$

- x $y * 1 = y$

Identity

Complement

* Distributes over +

Complement

Identity

Absorption theorem

- $x + x^*y = x$
- $x + x^*y =$
 - $x^*1 + x^*y$
 - $x^*(1 + y)$
 - x^*1
 - x
- $x^*(x + y) = x$
- $x^*(x + y) =$
 - $(x + 0)^*(x + y)$
 - $x + (0^*y)$
 - $x + 0$
 - x

Is $(\{0,1\}, +, \cdot)$ a Boolean Algebra?

Huntington Postulates

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- $\forall x \in B$, there exists a complement $x' \in B$ such that
 $x + x' = 1$ and $x * x' = 0$

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

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0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
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 $x + x' = 1$ and $x \cdot x' = 0$

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

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- $\forall x \in B$, there exists a complement $x' \in B$ such that
 $x + x' = 1$ and $x \cdot x' = 0$

x	y	z	$y + z$	$x \cdot (y + z)$
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

$x \cdot y$	$x \cdot z$	$(x \cdot y) + (x \cdot z)$
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	1	1
1	0	1
1	1	1

Is $(\{0,1\}, +, \cdot)$ a Boolean Algebra?

Huntington Postulates

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- $\forall x \in B$, there exists a complement $x' \in B$ such that $x + x' = 1$ and $x * x' = 0$

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

x	x'
0	1
1	0

Proving by truth tables

Absorption Theorem

- $x + x^*y = x$
- Do both sides yield identical results for all possible combinations of the variables involved?

<i>x</i>	<i>y</i>
0	0
0	1
1	0
1	1

<i>xy</i>	<i>x + xy</i>
0	0
0	0
0	1
1	1

Proving by truth tables

DeMorgan's Law

- $(x+y)' = x'y'$
- Do both sides yield identical results for all possible combinations of the variables involved?

x	y	$x + y$	$(x + y)'$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

x'	y'	$x'y'$
1	1	1
1	0	0
0	1	0
0	0	0

Cancellation law?

- If $x+z = y+z$ then $x=y$
- Holds in real numbers, $+$, $*$. But not in Boolean Algebra as $0+1=1+1$, but $0 \neq 1$

Big Idea

Switching algebra

- “One of the most important masters thesis ever written.. a landmark in that it helped to change digital circuit design from an art to science”

A Symbolic Analysis of Relay and Switching Circuits*

*Claude E. Shannon***

I. Introduction

In the control and protective circuits of complex electrical systems it is frequently necessary to make intricate interconnections of relay contacts and switches. Examples of these circuits occur in automatic telephone exchanges, industrial motor-control equipment, and in almost any circuits designed to perform complex operations automatically. In this paper a mathematical analysis of certain of the properties of such networks will be made. Particular attention will be given to the problem of network synthesis. Given certain characteristics, it is required to find a circuit incorporating these characteristics. The solution of this type of problem is not unique and methods of finding those particular circuits requiring the least number of relay contacts and switch blades will be studied. Methods will also be described for finding any number of circuits equivalent to a given circuit in all operating characteristics. It will be shown that several of the well-known theorems on impedance networks have roughly analogous theorems in relay circuits. Notable among these are the delta-wye and star-mesh transformations, and the duality theorem.

The method of attack on these problems may be described briefly as follows: any circuit is represented by a set of equations, the terms of the equations corresponding to the various relays and switches in the circuit. A calculus is developed for manipulating these equations by simple mathematical processes, most of which are similar to ordinary algebraic algorithms. This calculus is shown to be exactly analogous to the calculus of propositions used in the symbolic study of logic. For the synthesis problem the desired characteristics are first written as a system of equations, and the equations are then manipulated into the form representing the simplest circuit. The circuit may then be immediately drawn from the equations. By this method it is always possible to find the simplest circuit containing only series and parallel connections, and in some cases the simplest circuit containing any type of connection.

Point of view is worth 80 IQ points — Alan Kay

What must be the square which, when increased by ten of its own roots, amounts to thirty-nine?

Halve the number of roots, which in the present instance yields five. Multiply this by itself; the product is twenty-five. Add this to thirty-nine; the sum is sixty-four. Now take the root of this, which is eight, and subtract from it half the number of roots, which is five; the remainder is three. This is the root of the square which you sought, and the square itself is nine. — Al Khwarizmi (830 CE)

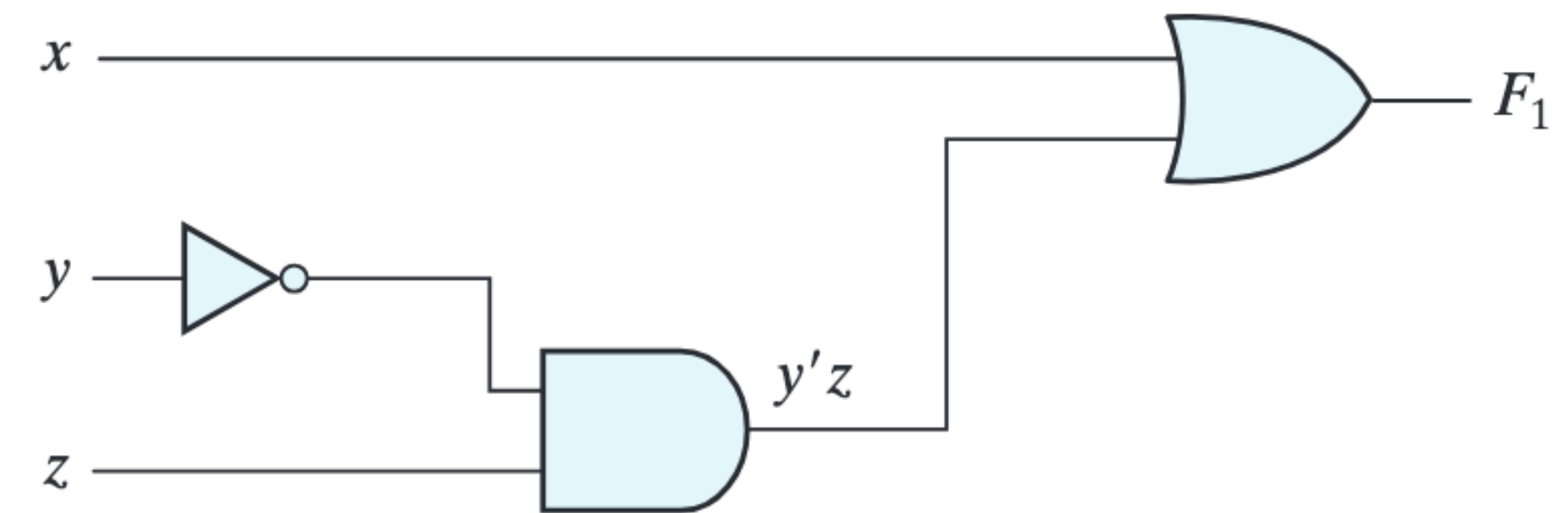
$$* x^2 + 10x = 39$$

$$* x = \frac{-10 \pm \sqrt{10^2 + 4 * 39}}{2}$$

$$* x = \frac{-10 + 16}{2}$$

Schematic

- Circuit connects input wires to output wires
- Can think of it as a boolean function: variables of the function are inputs and the function result is the output
- $F_1 = x + y'z$



Boolean Function

- Algebraic expression consisting of binary variables, constants 0 and 1, and logic operation symbols.
- For a given value of input, the output is either 0 or 1.
 - $F_1 = x + y'z$.
1 when $x=1$ or when $yz=01$

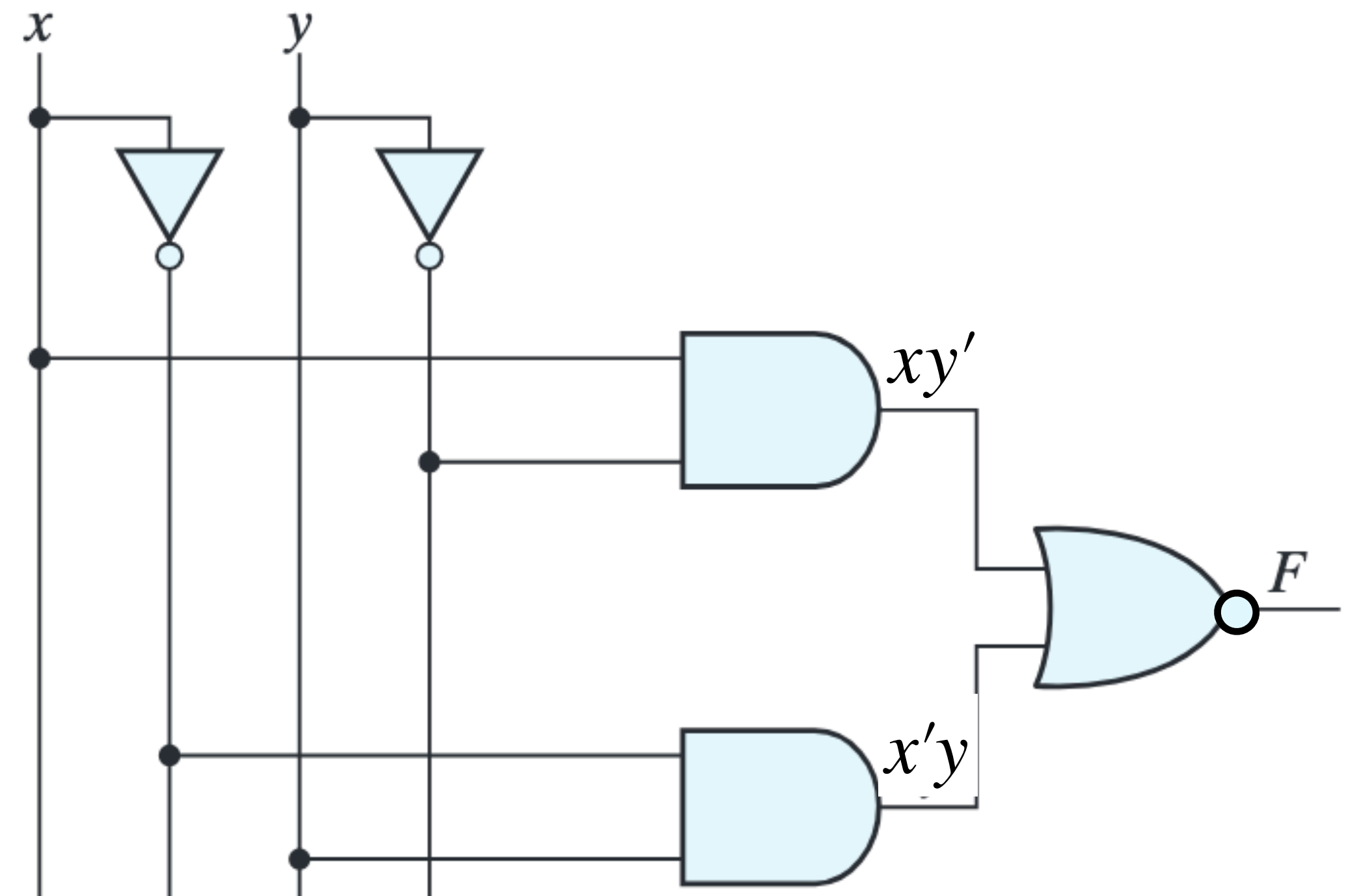
x	y	z	F_1
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Complement of a function

- Complement of a function F' : interchange 1 to 0 and 0 to 1 in the output of F .
- DeMorgan's law:
 - $(x + y + z)' = x'y'z'$
 - $(xyz)' = x' + y' + z'$
- $F_1 = x'yz' + x'y'z$
 - $F'_1 = (x + y' + z)(x + y + z')$

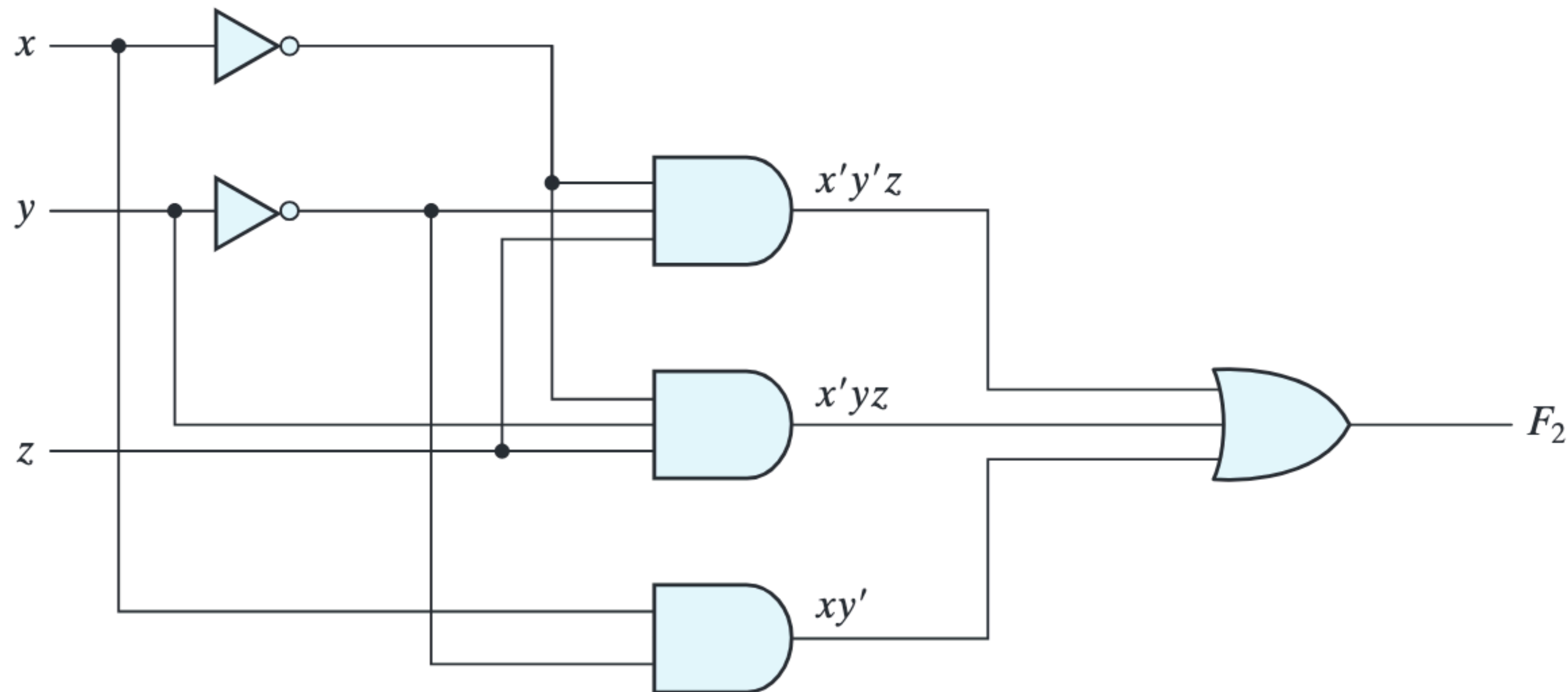
Complement of a function (2)

- $F' = (x'y + xy')'$
 - $(x'y)'(xy')'$
 - $(x+y')(x'+y)$
 - $xx' + xy + y'x' + y'y$
 - $xy + x'y'$



Truth tables and logic diagrams

- $F_2 = x'y'z + x'yz + xy'$



(a) $F_2 = x'y'z + x'yz + xy'$

x	y	z	F_2
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Simpler expressions lead to simpler circuits

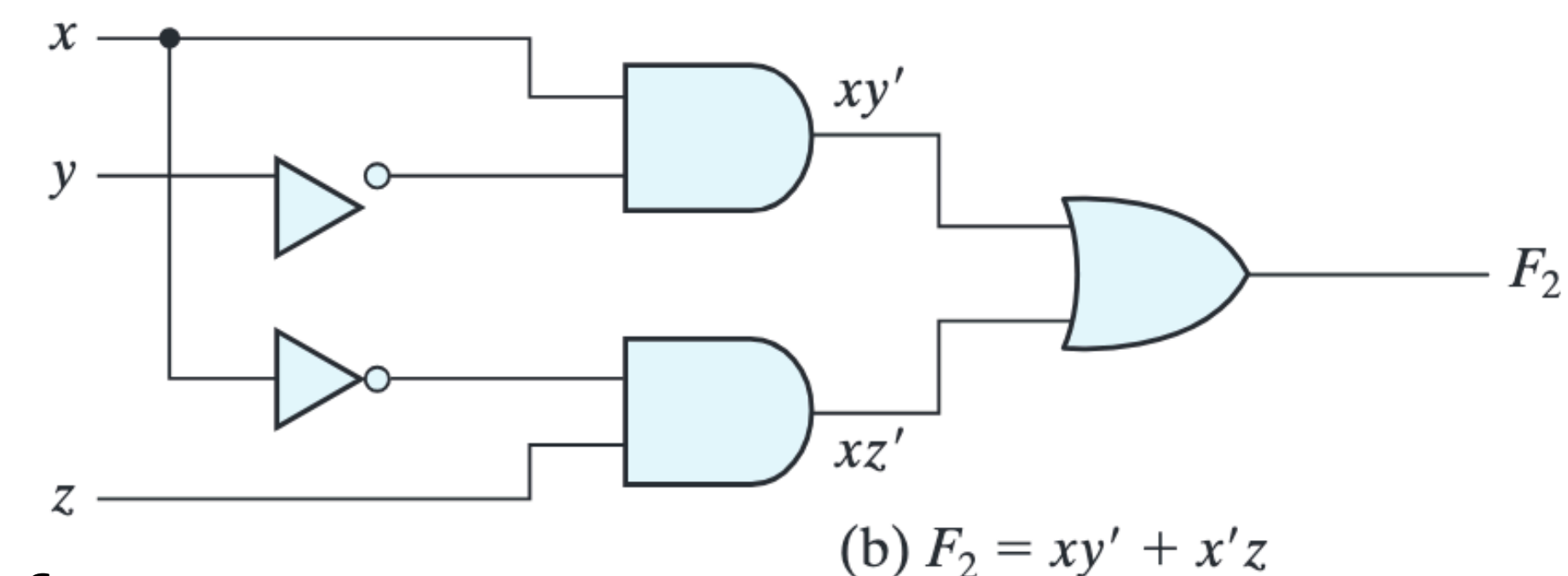
- There is only one way in which a function can be expressed in the truth table
 - But in a variety of ways in its algebraic and schematic form, all of which have equivalent logic

x	y	z	F_2
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

- $F_2 = x'y'z + x'yz + xy'$

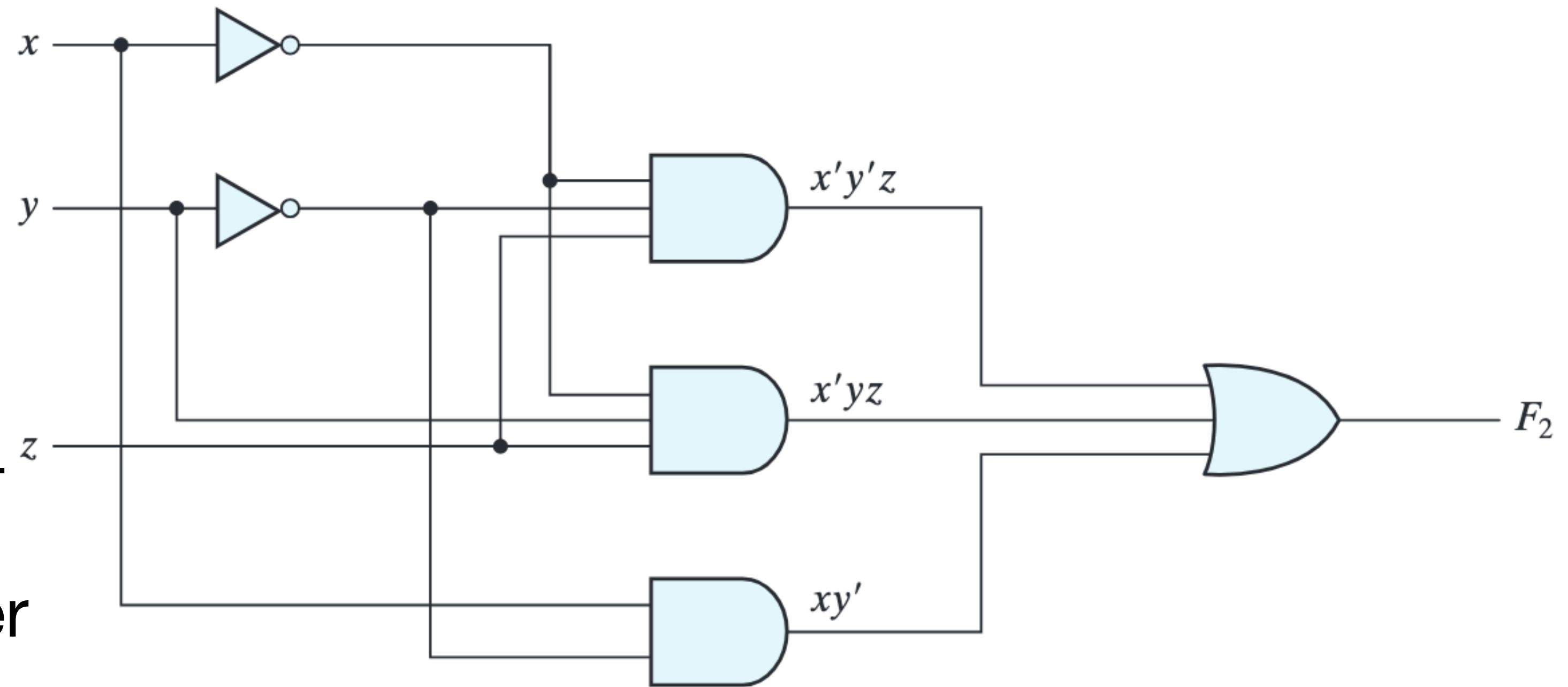
- $F_2 = x'z + xy'$

- Both produce the same truth table: same outputs for all possible input combinations. Both are equivalent

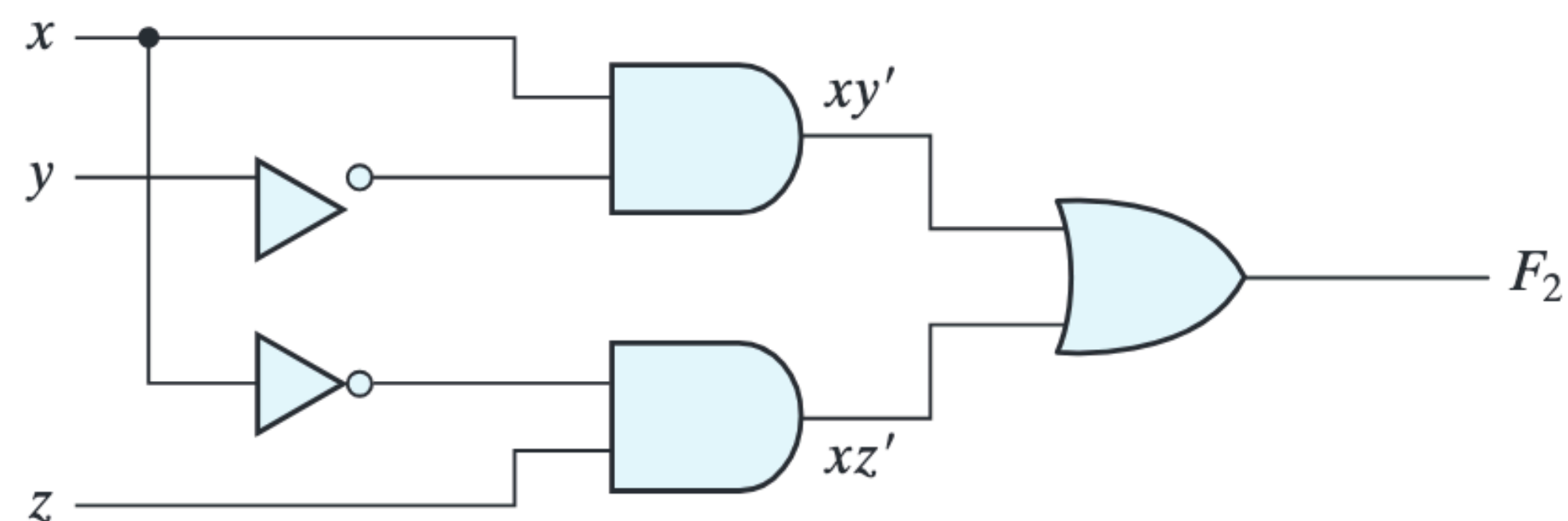


Algebraic Manipulations

- $F_2 = x'y'z + x'yz + xy'$. 3 terms and 8 literals.
- $F_2 = x'z + xy'$. 2 terms and 4 literals. Preferable because it requires fewer wires and fewer gates



(a) $F_2 = x'y'z + x'yz + xy'$



(b) $F_2 = xy' + x'z$

Simplify the following Boolean expressions

- $x(x' + y)$
 - xy
- $x + x'y$
 - $x + y$
- $(x + y)(x + y')$
 - x
- $xy + x'z + yz$
 - $xy + x'z + yz(x + x')$
 - $x(y + yz) + x'(z + yz)$
 - $xy + x'z$